

EMC used its AR and virtual reality (VR) apps to visually represent the capabilities of its converged and hyper-converged solutions in IT applications. Attendees got a much clearer perspective of the scopes and practical use cases of the offerings through AR, compared to normal text or video demonstrations.

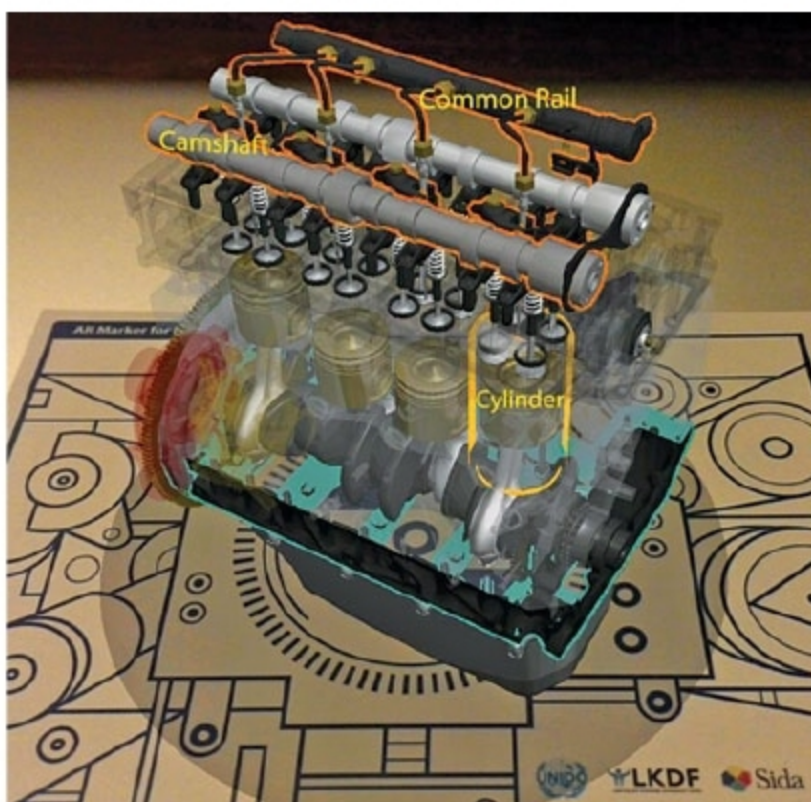
Retailers are also leveraging this technology. One of the early adaptors of AR, US-based furniture e-store WayFair launched WayFair View app, which lets customers digitally try out furniture placements in their home, allowing them to make purchase decisions with ease. The move has brought about substantial growth in the business for the company.

Following suite, global players like Ikea and Zara, and Indian e-tailers like Myntra and Jabong, are also curating their strategies for employing AR fully. This will allow users to try out products like clothes and accessories by creating their digital *avatars*.

A 2017 report by Boston Consulting Group estimated six to ten per cent increase in revenue from such technologies leading to personalisation of customer shopping, which includes AR.

Training and skill development. Efficient training of employees does not only come by verbal knowledge transfer or taking notes. Employees need to get hands-on practice. At jobs where training workers on the production floor or field may be too risky—but they still need to receive practical training—AR systems become the best bet. Industries and factories, medical institutions, airline ground crew, educational bodies and even corporate offices can use AR and VR models to train their workforce with necessary theoretical knowledge and assign practical tasks.

The United Nations Industrial Development Organization (UNIDO), a UN body that promotes industrial growth by skilling the underprivileged for employment, uses AR solutions for the purpose. For example, for a project in Africa—a continent where there is lack of basic education and skills, and language barrier was a challenge towards re-skilling—UNIDO invested in an AR platform to help the



AR training module for engine developed by Eon Reality
(Credit: eonreality.com)

candidates learn basic and advanced skills for training. The process helped in faster and more effective teaching by encouraging visual and hands-on learning.

Remote collaboration. AR platforms connected to the cloud can be the trump card for geographically-distributed enterprises to improve efficient collaboration between remote teams. Product design houses, manufacturing floors, on-field jobs in oil and gas or steel plants, and so on can make use of this feature. Development teams can share digital models of their progress in a project to let other members contribute remotely. Factory supervisors can provide on-field employees with visual guidance in an operation to ensure safe and efficient execution of a task. Essential notes and data of on-field systems can be accessed on-the-go so that workers can act accordingly.

Global automaker OPEL have adopted an AR solution by developers Eon Reality for factory workers in their product assembly process. The solution consists of an AR headset and a smartwatch. Workers receive real-time information and digital guidance on their headset about the operations they need to carry out. Accordingly, they control robots and machineries in the factory using their smartwatches. The process showed improvement in production efficiency, and reduced

errors and stress on workers. Such solutions can be very useful in labour-surplus countries like India, where assembly operations are often outsourced because of low labour costs.

There are many more ways AR can play crucial roles in transforming businesses. Its benefits are multifold, including much higher customer engagement, lower operational errors, better efficiency and higher productivity. AR also reduces financial expenses while cutting down on

training investments, reducing rates of returned orders, reducing risk of product damage during demonstrations and so on. The ROI comes quick through these benefits.

Costs and limitations

Cost for AR solutions for businesses is proportionate with the scale of business requirement. Some solutions may only come as software platforms and applications, and only require compatible smartphones (or tablets) as hardware. Other solutions may require separate hardware, which are mostly in the form of AR headsets. Prices vary widely from vendor to vendor. However, basic AR hardware often costs around ₹ 10,000 per unit (about US\$ 150), and comes with basic features for individual use. For enterprise-level usage or large-scale applications, complete AR systems may start from US\$ 2000 a year, with no fixed higher cap. Purchase models may also differ.

Apart from the upfront investment, businesses also have to ensure that data and media they would like to use for AR models are accurate. Users also require basic technical know-how to be able to handle AR hardware. As AR is a developing technology, users may experience glitches in some AR programs. Nonetheless, if used properly with the right plans in place, AR can be a highly beneficial tool towards success.